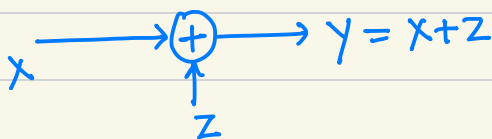


Additive channels



$$\frac{1}{n} \sum_{i=1}^n x_i^2 \leq P$$

z is independent of x

$$C = \max_{p_x: EX^2 \leq P} \mathcal{I}(x; y)$$

$$C = \frac{1}{2} \log \left(1 + \frac{P}{\sigma_z^2} \right) \quad \text{when } z \sim \mathcal{N}(0, \sigma_z^2)$$

Achievability proof:

Take p_x that maximizes $\mathcal{I}(x; y)$ s.t. $EX^2 \leq P - \epsilon$

Converse:

$$nR = H(W) = \mathcal{I}(W; \hat{W}) + H(W | \hat{W})$$

$$\leq \mathcal{I}(x^n; y^n) + n\epsilon_n \quad \text{where } \epsilon_n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$= \underline{h(y^n)} - \underline{h(y^n | x^n)} + n\epsilon_n$$

$$\leq \sum_{i=1}^n h(y_i) - \sum_{i=1}^n h(y_i | x_i) + n\epsilon_n$$

$$= \sum_{i=1}^n \underline{\mathcal{I}(x_i; y_i)} + n\epsilon_n$$

$$X^n = (x_1, \dots, x_n)$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$p_1 \quad p_2 \quad p_n$$

$$\left. \begin{aligned} h(y^n | x^n) &= h(x^n + z^n | x^n) \\ &= h(z^n | x^n) \\ &= h(z^n) \\ &= \sum_{i=1}^n h(z_i) \\ &= \sum_{i=1}^n h(y_i | x_i) \end{aligned} \right\}$$

$$\rightarrow \underline{p = \frac{1}{n} \sum_{i=1}^n p_i} \quad \underline{x \sim p}$$

$$I(x; y) \geq \frac{1}{n} \sum_{i=1}^n I(x_i; y_i)$$

$$\Rightarrow n I(x; y) \geq \sum_{i=1}^n I(x_i; y_i)$$

$$nR \leq n I(x; y) + n \epsilon_n$$

$$\Rightarrow R \leq \underline{I(x; y)} + \epsilon_n$$

$$\Rightarrow R \leq \underline{C} + \underbrace{\epsilon_n}_{\downarrow 0}$$

$$\Rightarrow R \leq C$$

p satisfies the power constraint
i.e. $EX^2 \leq P$

$$EX_1^2 = P_1$$

$$EX_2^2 = P_2$$

$\vdots$

$$EX_n^2 = P_n$$

$$\underline{EX^2} = \int x^2 p(x) dx$$

$$= \int x^2 \cdot \left(\frac{1}{n} \sum p_i(x) \right) dx$$

$$= \frac{1}{n} \sum_{i=1}^n \underbrace{\int x^2 p_i(x) dx}_{EX_i^2}$$

$$= \frac{1}{n} \sum_{i=1}^n EX_i^2 \leq P$$

$$\text{since } \frac{1}{n} \sum_{i=1}^n x_i^2 \leq P$$

Entropy power inequality (EPI)

X, Z - independent

$$Y = X + Z$$

$$\underline{\underline{2^{2h(y)} \geq 2^{2h(x)} + 2^{2h(z)}}} \quad \leftarrow$$

with "=" if x, z are Gaussian

Gaussian case:

$$x \sim \mathcal{N}(0, P) \quad , \quad z \sim \mathcal{N}(0, \sigma^2)$$

$$\underline{h(x) = \frac{1}{2} \log(2\pi e P)} \quad , \quad \underline{h(z) = \frac{1}{2} \log(2\pi e \sigma^2)}$$

$$y = x + z \sim \mathcal{N}(0, P + \sigma^2)$$

$$\underline{h(y) = \frac{1}{2} \log(2\pi e (P + \sigma^2))}$$

$$\begin{array}{l|l} 2^{2h(x)} = 2\pi e P & 2^{2h(z)} = 2\pi e \sigma^2 \\ \Rightarrow 2^{2h(x)} = 2\pi e P & \end{array}$$

$$2^{2h(y)} = 2\pi e (P + \sigma^2)$$

$$\begin{aligned} 2^{2h(y)} &= 2\pi e (P + \sigma^2) \\ &= 2\pi e P + 2\pi e \sigma^2 \\ &= 2^{2h(x)} + 2^{2h(z)} \end{aligned}$$

Among all additive channels, AWGN channel has the minimum capacity!

Diagram: $x \rightarrow (+) \xrightarrow{z} y = x + z$

$\frac{1}{n} \sum_{i=1}^n x_i^2 \leq P$; $\underline{\underline{Ez^2 \leq \sigma^2}}$

$C = \max_{Ex^2 \leq P} I(x; y)$

$Z \sim \mathcal{N}(0, \sigma^2)$ is the worst noise

$$C = \max_{Ex^2 \leq P} I(x; y)$$

$$\begin{aligned} & \frac{h(y|x)}{=} h(x+z|x) \\ & = h(z|x) \\ & = h(z) \end{aligned}$$

$$= \max_{Ex^2 \leq P} h(y) - h(z)$$

$$= \max_{Ex^2 \leq P} h(x+z) - h(z)$$

$$\geq \max_{Ex^2 \leq P} \frac{1}{2} \log \left[2^{2h(x)} + \underline{\underline{2^{2h(z)}}} \right] - \underline{h(z)}$$

$$= \frac{1}{2} \log \left[2^{2 \cdot \max_{\underline{\underline{Ex^2 \leq P}}} h(x)} + 2^{2h(z)} \right] - h(z)$$

$$= \frac{1}{2} \log \left[2^{2 \cdot \frac{1}{2} \log(2\pi e P)} + 2^{2h(z)} \right] - h(z) \quad Ex^2 \leq P$$

$$= \frac{1}{2} \log \left[2\pi e P + 2^{2h(z)} \right] - \underline{\underline{h(z)}}$$

$$= \frac{1}{2} \log \left[\frac{2\pi e P}{2^{2h(z)}} + 1 \right]$$

$$\geq \frac{1}{2} \log \left[\frac{2\pi e P}{\cancel{2\pi e \sigma^2}} + 1 \right]$$

$$= \frac{1}{2} \log \left(1 + \frac{P}{\sigma^2} \right)$$

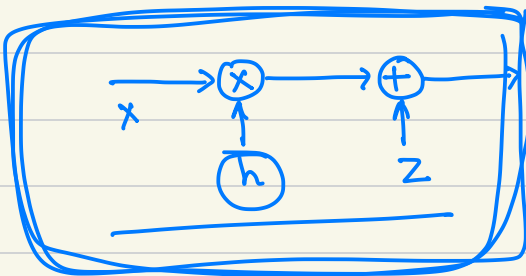
$$\frac{1}{2} \log 2^{2h(z)}$$

$$h(z) \leq \frac{1}{2} \log(2\pi e \sigma^2)$$

$$2h(z) \leq \log(2\pi e \sigma^2)$$

$$\Rightarrow \frac{2^{2h(z)}}{2} \leq 2\pi e \sigma^2$$

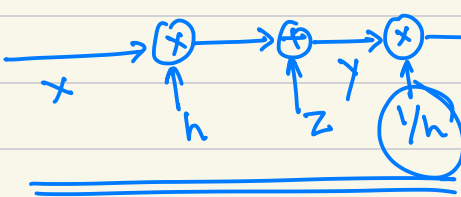
AWGN channel with multiplicative gain:



$$y = hx + z$$

$$\frac{1}{n} \sum x_i^2 \leq P$$

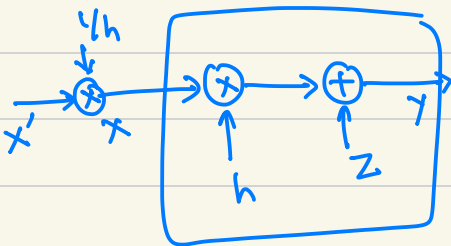
$$z \sim \mathcal{N}(0, \sigma^2)$$



$$y' = x + \frac{1}{h} z$$

$$= x + z' ; z' \sim \mathcal{N}(0, \frac{\sigma^2}{h^2})$$

$$\text{AWGN}(P, \frac{\sigma^2}{h^2})$$



$$y = x' + z$$

$$x' = hx$$

$$\Rightarrow E x'^2 = h^2 E x^2$$

$$E x^2 \leq P \Leftrightarrow \underline{E x'^2} \leq \underline{h^2 P}$$

$$C = \frac{1}{2} \log \left(1 + \frac{P h^2}{\sigma^2} \right)$$

$$z \sim \mathcal{N}(0, \sigma^2)$$

$$AWGN(h^2 P, \sigma^2)$$

$$SNR = \frac{h^2 P}{\sigma^2}$$

$$C = \frac{1}{2} \log \left(1 + \frac{P h^2}{\sigma^2} \right)$$